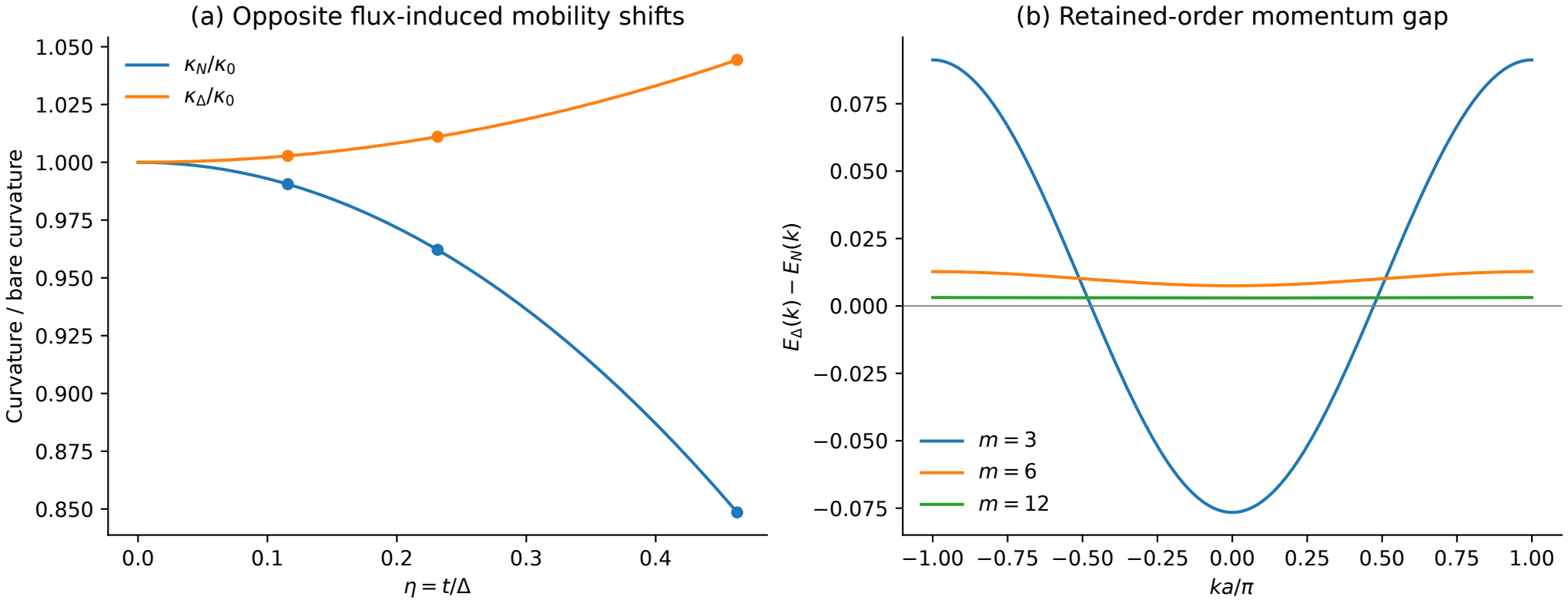
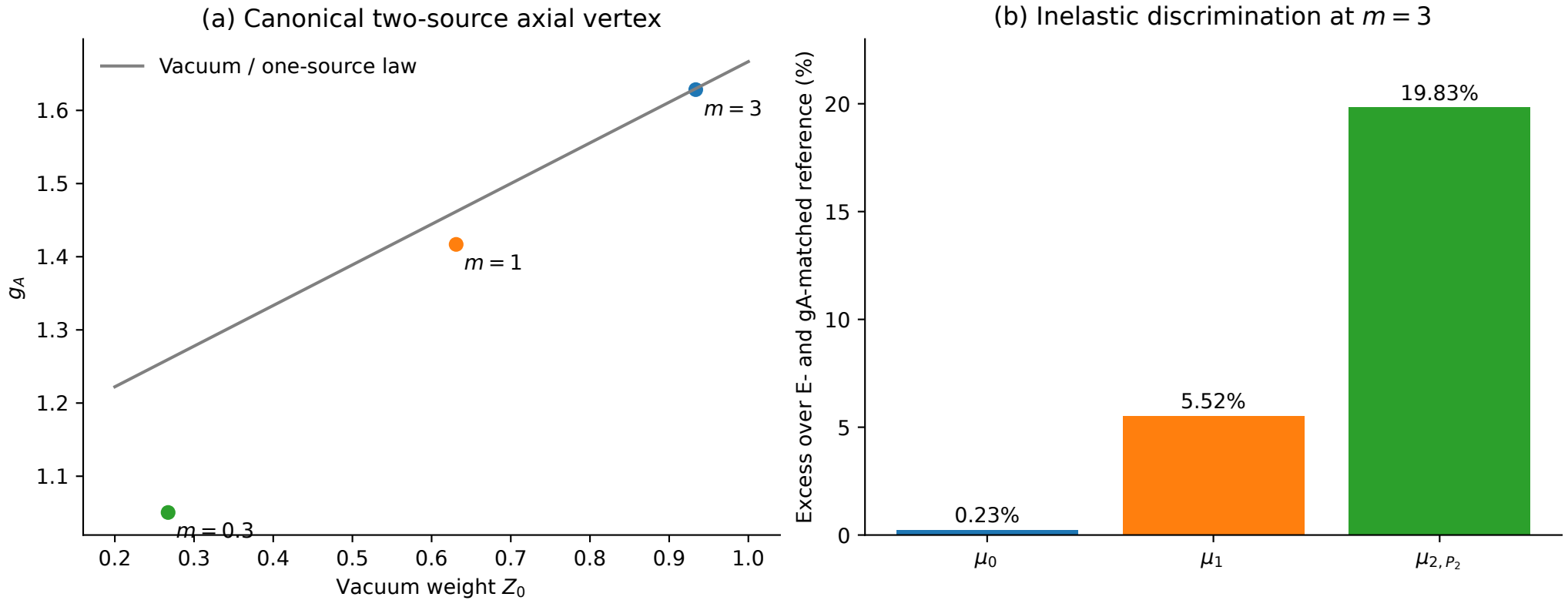
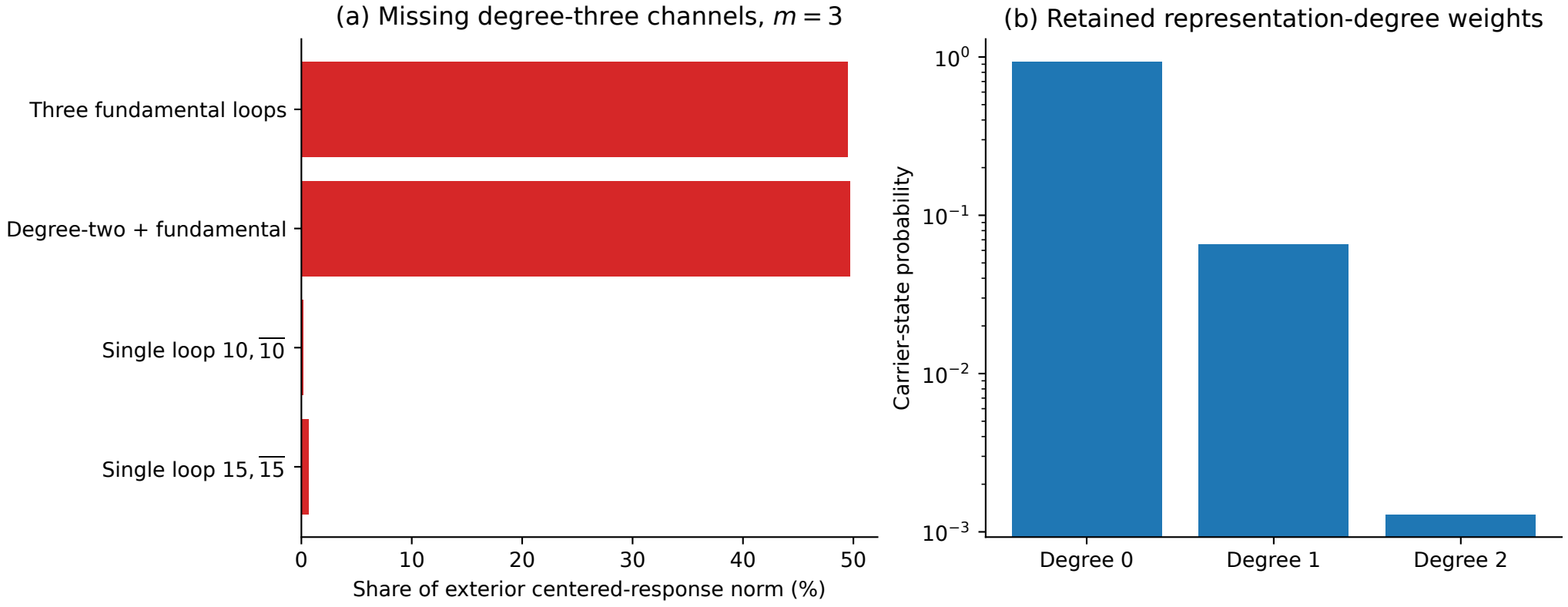

(a) Opposite flux-induced mobility shifts
Curvature / bare curvature
κ_N/κ_0
κ_Δ/κ_0
1.050
1.025
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.0
0.1
0.2
0.3
0.4
$\eta = t/\Delta$
(b) Retained-order momentum gap
$E_\Delta(k) - E_N(k)$
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
$m = 3$
$m = 6$
$m = 12$
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
ka/π


(a) Canonical two-source axial vertex
Vacuum / one-source law
$m = 3$
$m = 1$
$m = 0.3$
g_A
1.6
1.5
1.4
1.3
1.2
1.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Vacuum weight Z_0
(b) Inelastic discrimination at $m = 3$
Excess over E- and gA-matched reference (%)
20
15
10
5
0
19.83%
5.52%
0.23%
μ_0
μ_1
μ_{2,P_2}


(a) Missing degree-three channels, $m = 3$
Three fundamental loops
Degree-two + fundamental
Single loop $10, \overline{10}$
Single loop $15, \overline{15}$
0
10
20
30
40
50
Share of exterior centered-response norm (%)
(b) Retained representation-degree weights
Carrier-state probability
10^{0}
10^{-1}
10^{-2}
10^{-3}
Degree 0
Degree 1
Degree 2